CAPSTONE PROJECT FINAL REPORT

Data-Driven Approach to Analyze Amazon Product Pricing, Ratings, and Sentiment

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Introduction

In our Capstone Project, "Analyzing Amazon Product pricing, Ratings and Sentiment," we analyze and predict key aspects of Amazon product data. Our initial plan was to do sentiment analysis, but as we dived deeper into the data, we changed our direction to three main goals because of limited data availability: (1) predicting product price range based on category features reviews and ratings; (2) understanding the elements that influence average review ratings; and (3) Can we accurately predict the sentiment of a review based on its text and other features? The dataset has over 10,000 records representing 894 features derived from Amazon products opens dataset. Using regression models and clustering methods, we will analyze data to help sellers make competitive pricing decisions and improve product quality while examining market dynamics.

For our Q2, to understand the elements that influence average review ratings, we applied a regression model to examine how different factors, including pricing, product category, and customer contacts, affect review scores to explain the components that affect average review ratings. The algorithm highlighted important connections between numerical attributes and product and revealed which features are most important in predicting review scores. However, given the accuracy and the findings, we decided a thorough investigation is required to measure the strength of these associations, even if this first research assisted in establishing predicting patterns. In order to have a better understanding of the influence of each feature, we will perform a correlation study to quantify the direct impact of each feature on review ratings.

Data Cleanup

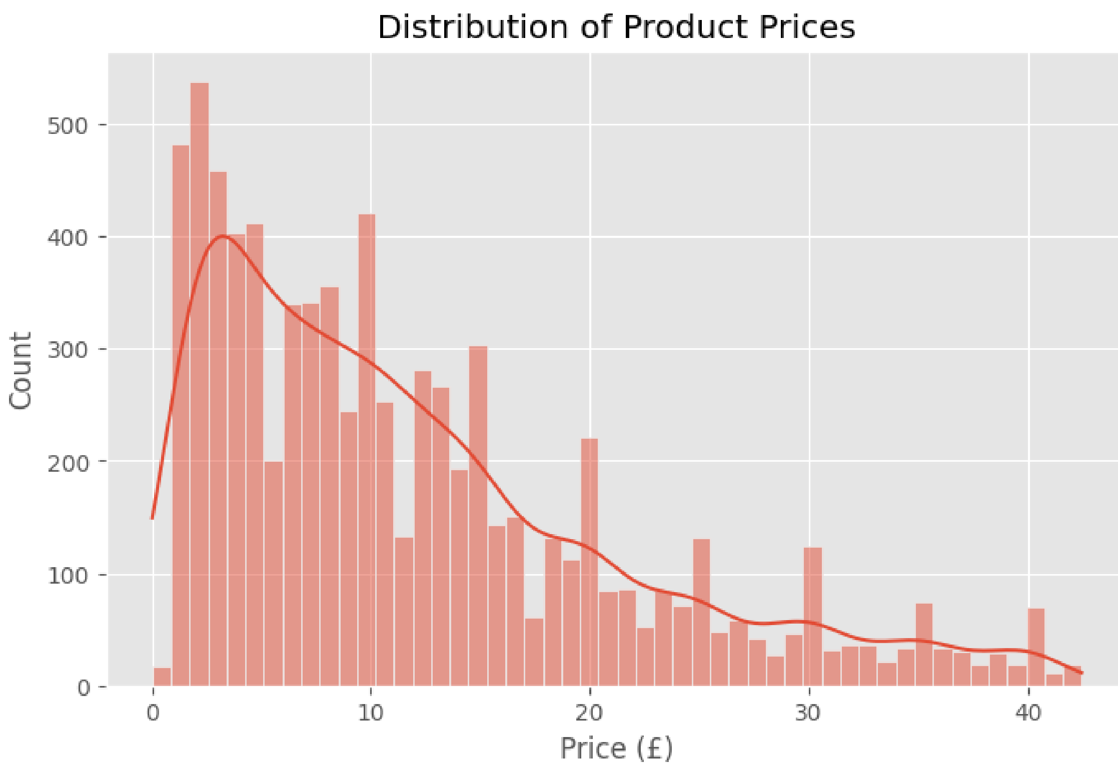
Addressing missing numbers, fixing errors, and structuring the data for additional analysis are the main goals of the first data cleaning step. To preserve data quality, missing values are first examined, and columns with more than 70% missing data are removed. Unwanted entries (such as "}") are discovered through further research and are also eliminated, along with rows that lack unique IDs. To guarantee consistency, the dataset's categorical variables—such as item category and stock availability—are scrutinized. To facilitate numerical analysis, pricing values are also cleaned by eliminating commas and currency symbols. Similar to this, review ratings and product availability are transformed into numerical representations so that they can be used in modelling.

Even though this data cleaning procedure fixes a lot of problems with data quality, it can still be improved. For numerical values, imputation techniques (mean, median, or mode) could be taken into consideration rather than just removing columns with missing data. The most common category may be used to fill in missing entries for categorical data. To find and deal with inaccurate results that can distort analysis, outlier identification should also be carried out. Data usability can also be improved by standardizing column names and making sure that data types are the same for comparable variables. The dataset can be better prepared for precise predictive modelling and insightful analysis by improving these procedures.

Exploratory Data Analysis

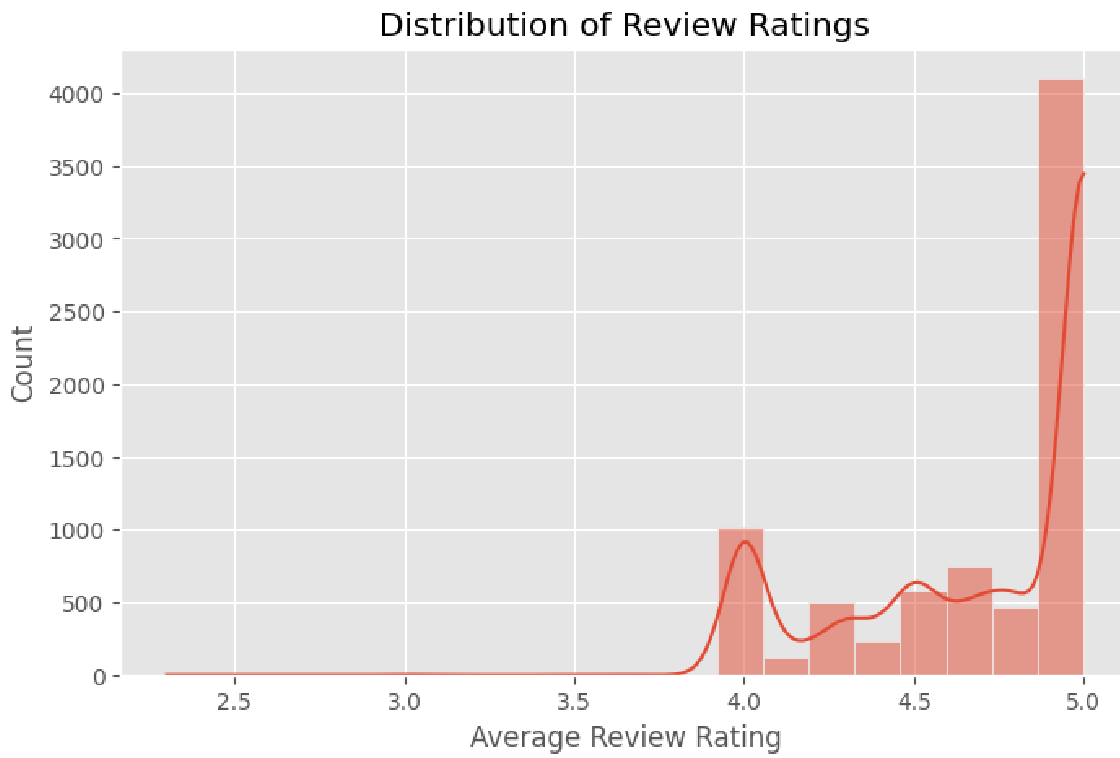
The dataset structure was explored through an EDA to identify missing values and reveal key trends before starting predictive modeling.

1.**Price Distribution**



The product price distribution shows the right skew which means many products cost less while there are some products with high prices. The high-priced products might belong to specific brands or the high-end market.

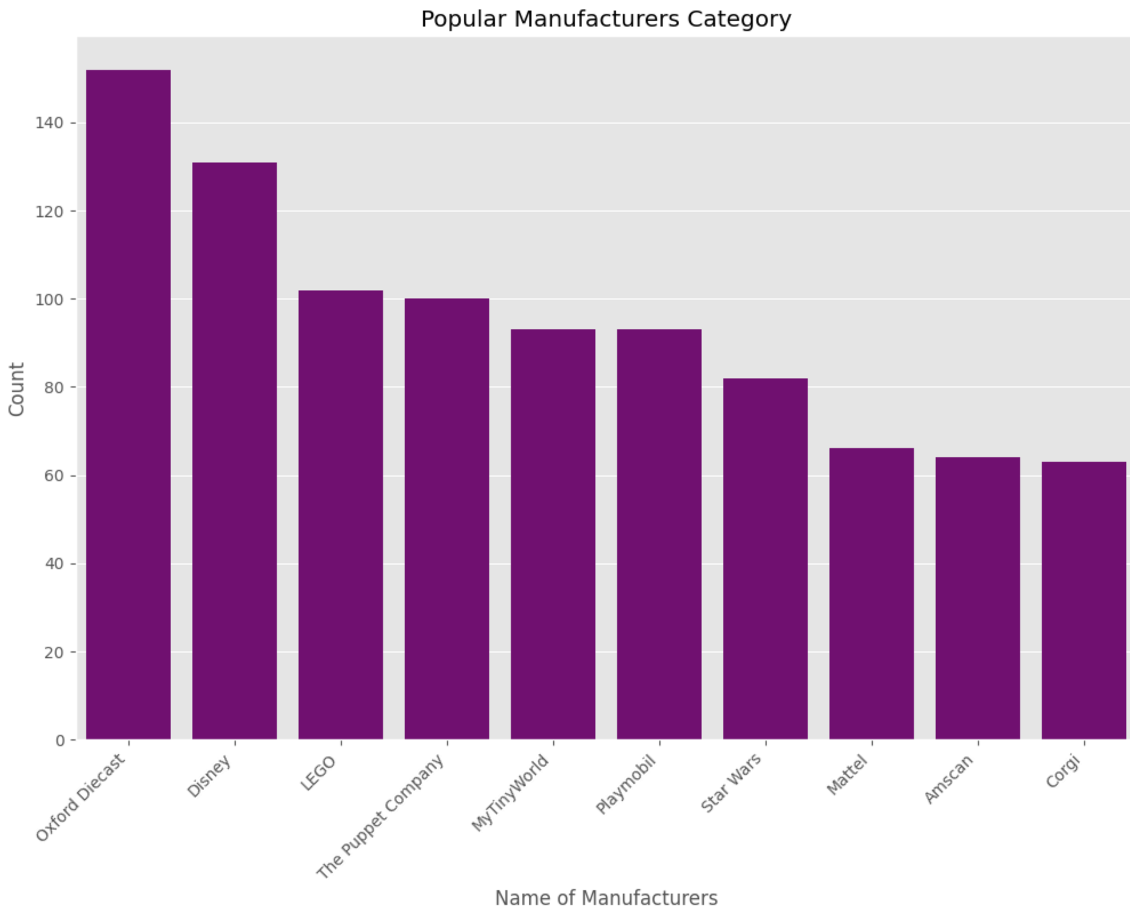
2. **Average Review Rating**



Most products received ratings between 4 and 5 stars, which indicates positive feedback dominates the review data.

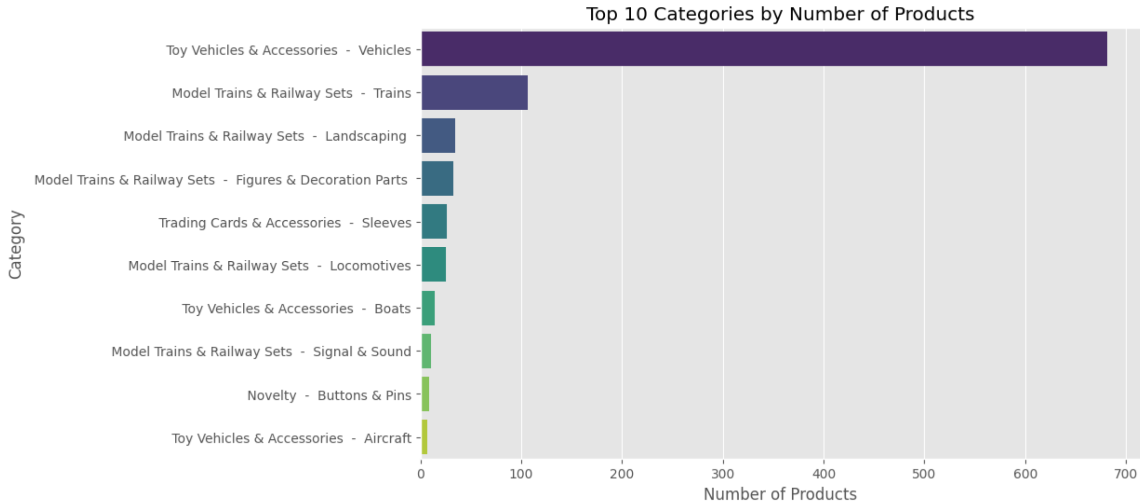
3.**Number of Reviews**

	manufacturer	average_review_rating(out of 5)	count
1345	Oxford Diecast	4.790789	152
510	Disney	4.683206	131
1051	LEGO	4.715686	102
1847	The Puppet Company	4.757000	100
1276	MyTinyWorld	4.923656	93
1441	Playmobil	4.726667	93
1704	Star Wars	4.760976	82
1174	Mattel	4.671212	66
74	Amscan	4.714062	64
405	Corgi	4.784127	63



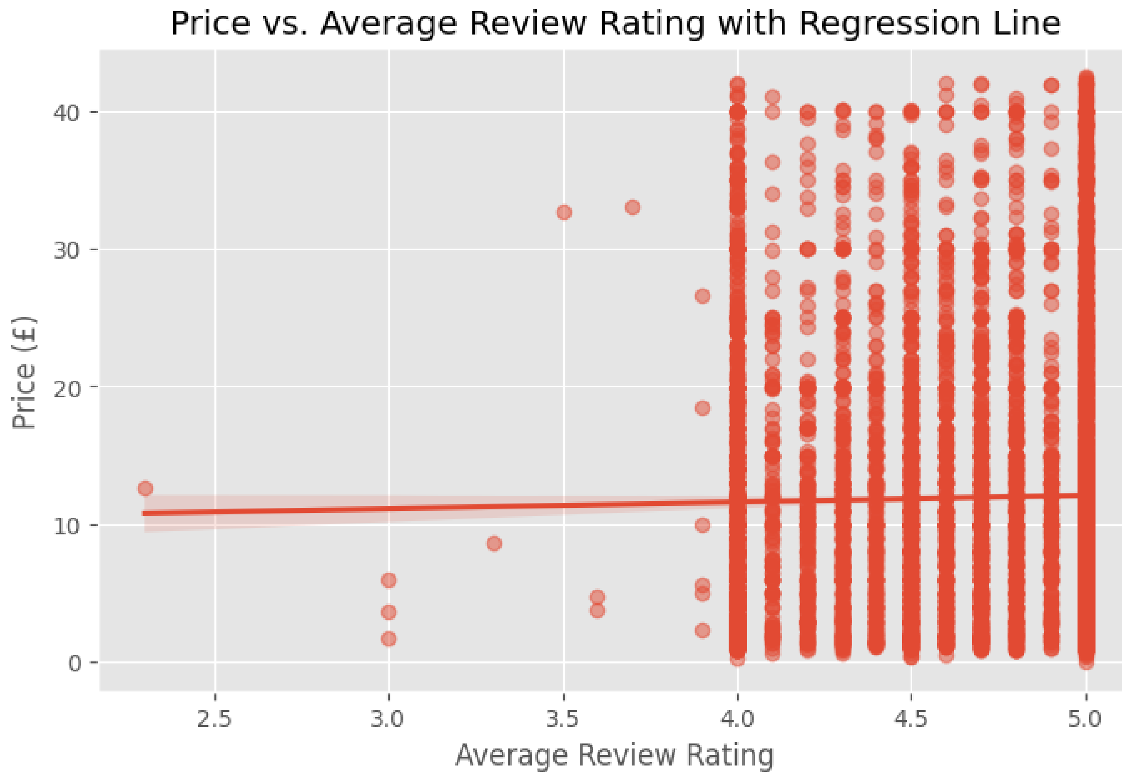
We analyzed the Popular Manufacturers Category and found that Oxford Diecast (which is the first time I've heard of them) might be a major market leader in the model area. They offer a wide range of products and have high sales, and this manufacturer also holds the second-highest rating among the top ten. While I understand that Disney and LEGO are well-known brands with high market acceptance, the fact that Oxford Diecast outperforms both indicates that they are extremely strong.

4. **Category Distribution**



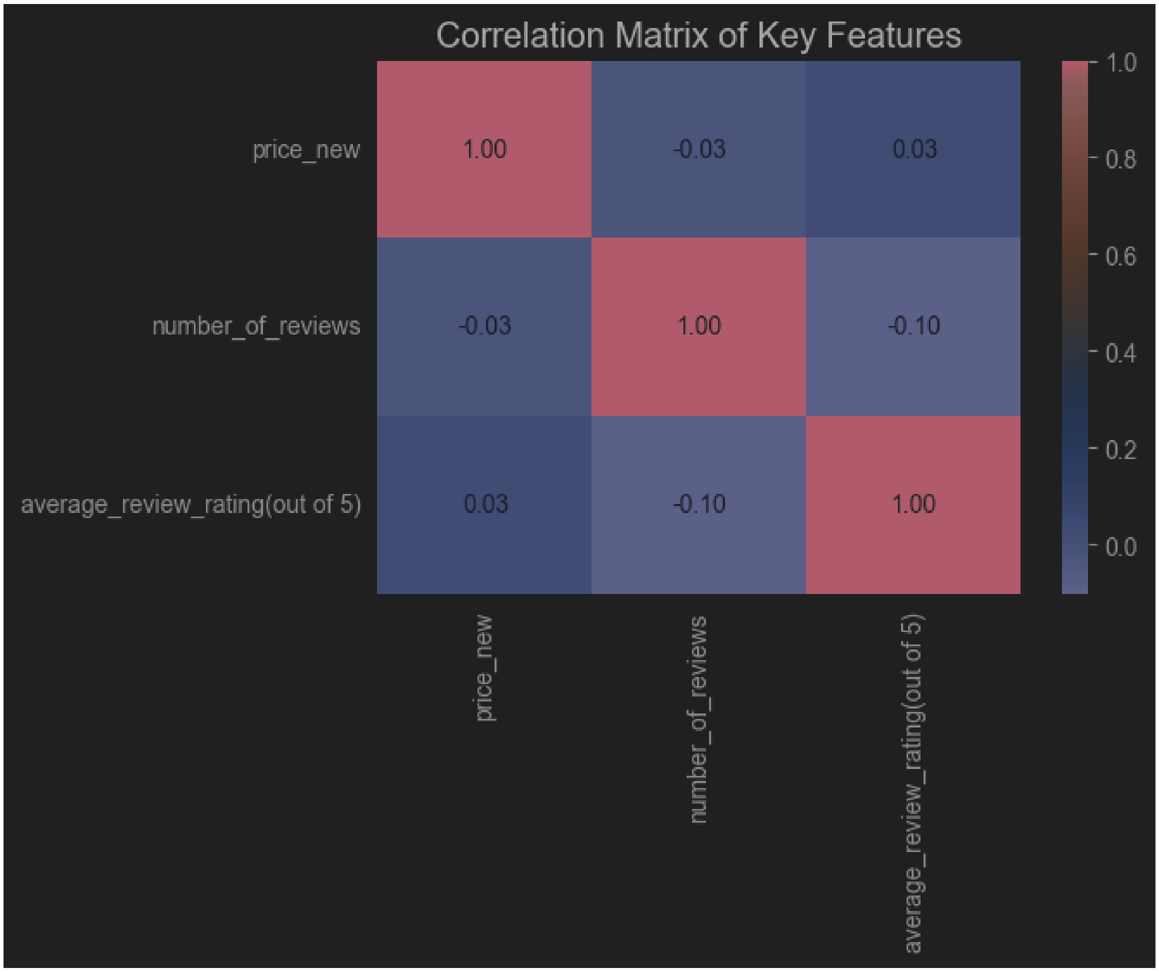
This is a "Top 10 Categories by Number of Products". To avoid readers, recognize the "vehicle" to the real cars, I put a category before the product. So, the Bar char shows, Toy Vehicles is the primary product category in the market, with a high consumer demand and a greater number of suppliers. Additionally, "Toy Vehicles & Accessories," as a broader category, it also includes Boats and Aircraft, which indicates that the diversity of products in this category is very high.

5. **Price vs. Review Rating**



Price vs. Average Review Rating with Regression Line: The regression line is nearly horizontal, indicating that there is almost no clear linear correlation between Price and Average Review Rating. This suggests that ratings have little influence on price, possibly because highly rated products are not necessarily more expensive, and low-rated products are not necessarily cheaper.

6. **Correlation Analysis**



A heatmap analysis of key numerical variables (price, number_of_reviews, average_review_rating) indicates: Review rating shows only a weak relationship with product price. The data shows a moderate positive link between product price and review count which implies that products that cost more tend to receive more reviews

Models

1. Logistical Regression Model for Q2

The summary below shows the improvements made to the Logistic Regression model that classifies products according to the number of reviews given in the dataset. Issues with feature selection, class imbalance, and inconsistent data created problems for the original model. We were able to increase the accuracy to 63% with improved feature engineering, model balancing, and data preparation, yielding more trustworthy predictions.

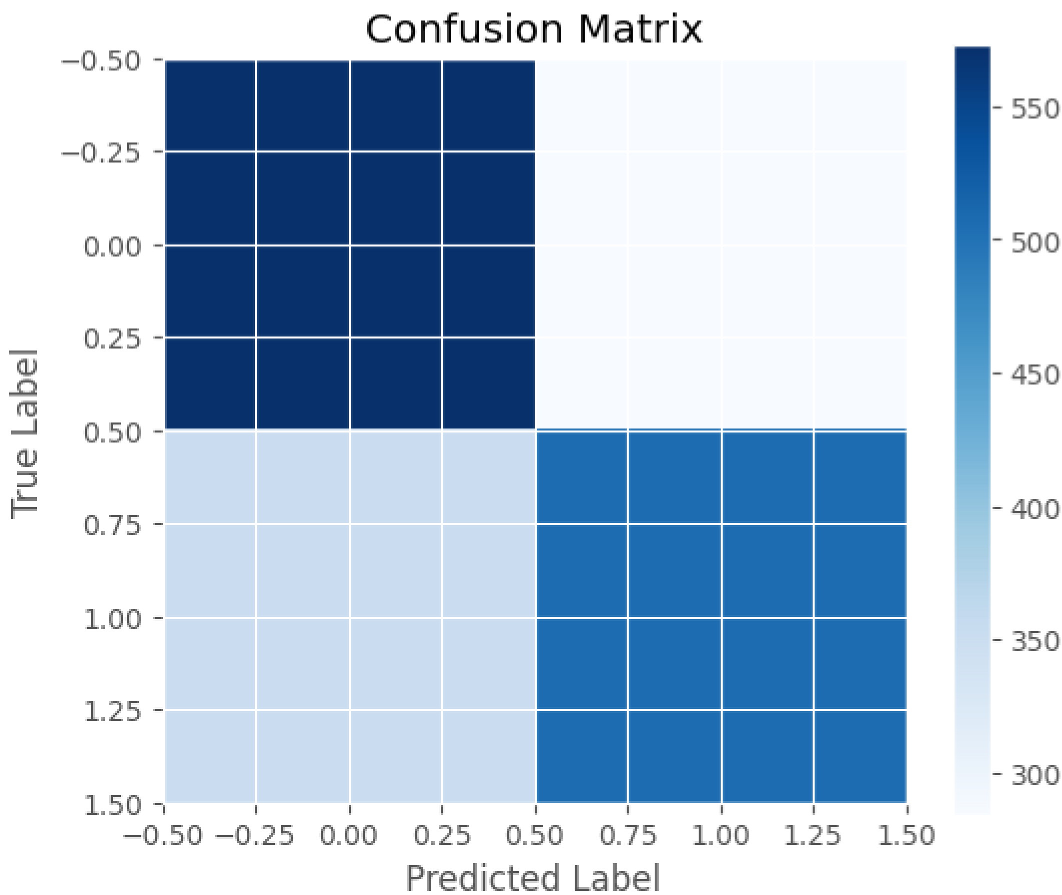
Enhancements to the Model

Cleaning and preparing the dataset was the first stage in model optimization. Non-numeric values, including price ranges and currency symbols (£), were extracted from the price column and averages for ranges were calculated. To ensure compatibility with Logistic Regression, numerical features were normalized using MinMaxScaler. To improve model performance, categorical variables such as manufacturer and product category were also transformed into numerical values using Label Encoding.

We also implemented feature selection and data balancing techniques to increase accuracy even more. New text-based features, such as number_of_answered_questions, which indicated customer interaction, were added to the dataset, which had previously concentrated on numerical properties. We used SMOTE (Synthetic Minority Oversampling Technique) to create synthetic samples to balance the dataset and reduce bias, as products with high ratings were underrepresented. Additionally, since terms like "Best Seller" or "Limited Edition" frequently indicated popularity, bi-grams and trigrams (n-grams (1,3)) were added to the text analysis to extract helpful context from product names.

***Visual* Analysis and Insights**

To understand the model's performance better, we created a confusion matrix that highlighted the model's weak points. Medium-reviewed products were most frequently misclassified, most likely due to feature similarities across high- and low-review categories. We evaluated the classification confidence using a multi-class ROC curve, which showed that medium-review products were still the hardest to categorize, while high- and low-review categories were more clearly separated.



Analysis and Results

Overall, the optimization method increased classification accuracy from the original baseline to 63% by correcting class imbalance, increasing feature selection, and improving data quality. However, even though the model is now more accurate, it might still be improved by evaluating more sophisticated machine learning models that can capture intricate correlations, like Random Forest, XGBoost, or Neural Networks. It would provide deeper insights into the popularity of a product that may be possible with additional text-based capabilities, such as sentiment analysis from user evaluations. Lastly, the model's classification performance could be further improved by hyperparameter tuning, especially when it comes to distinguishing medium-review products.

We believe that the accuracy and classification confidence of the model can be further enhanced beyond 63% by putting these extra improvements into practice, which will make it even more useful for classifying product reviews. Additionally, to respond to our Q2, we'll run a correlation study for more accurate results.

Random Forest

Introduction

This section will explain how improvements were applied and the reason when we switched model from Naïve Bayes to random forest. With this upgrade, the new random forest model achieved an 83.8% accuracy rate which significantly surpasses the previous result of 62%.

Model Upgrades and Justifications

There are some adjustments ***that were*** implemented to boost the predictive capabilities of the model. First, Random Forest classifier became an essential enhancement because it handles non-linear relationships well, minimizes overfitting through ensemble learning methods, and processes numerical and categorical data efficiently. Random Forest can detect complex feature interactions which Bayes cannot because Bayes assumes features are independent. The approach reveals how features contribute to model decisions which boosts model interpretability.

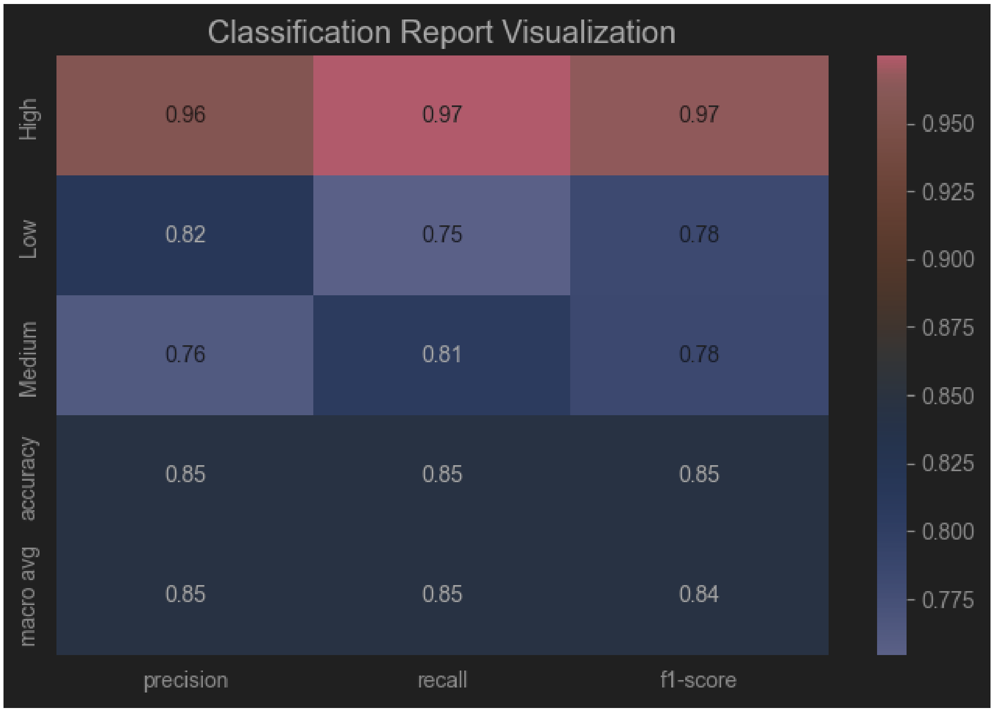
The feature engineering process was significantly improved. The TF-IDF representation expanded to capture deeper contextual meaning from textual data by increasing maximum features to 2000 and extending the ngram range to (1,3). The text features received enhancements through the ‘customer_questions_and_answers’ and the creation of extra features including ‘num_questions’, ‘text_length’, ‘avg_word_length’, and ‘unique_word_ratio’.

Numerical attributes received better handling through MinMaxScaler normalization while categorical features were transformed into a compatible encoding for the model.

The random forest has a significant advancement by implementing SMOTE which created synthetic samples for minority classes to achieve dataset balance. The data distribution is ***heavily focused*** on the medium price category required intervention because such imbalance risks introducing classification bias. To preserve class proportions between training and test data sets while enhancing stability, we implemented Stratified K-Fold Cross-Validation with five splits.

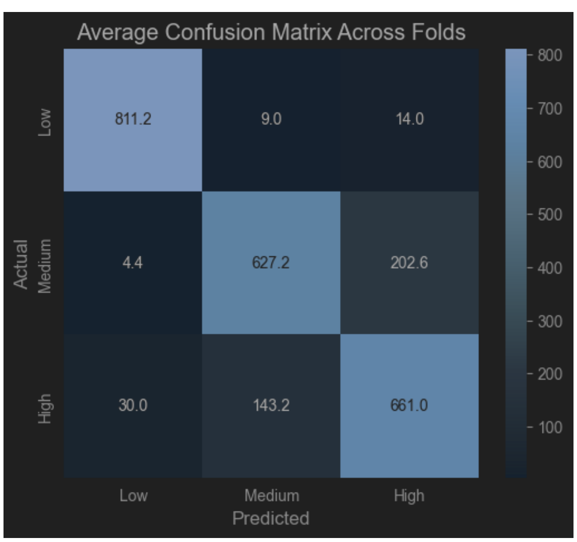
Analysis and Results

The improved model gains a significant substantial performance. The cross-validation accuracy scores showed consistent predictions while remaining between 82.9% and 84.5% with a mean accuracy of 83.8% and a standard deviation of 0.56%. The classification report validated excellent performance results for each price category. The High price segment demonstrated precision and recall scores of 0.96 and 0.97 respectively which validated the model's precise identification of expensive products. The recall for the Low-price category at 0.75 indicates that some products were incorrectly classified as medium price.



Visualizations

The confusion matrix averaging over five cross-validation folds revealed high prediction accuracy with minimal errors between Low and Medium price ranges. The analysis revealed a minor pattern of medium products being incorrectly categorized as High which provides a chance to enhance the model's precision.



SVM and XGBoost Model for Question 3:

With our present model, we can examine whether review sentiment varies significantly across product categories or price points. Nevertheless, the category column is absent from our dataset, which restricts our capacity to compare sentiment among various product categories. We would categorize things and determine their average sentiment scores if we could get this information. To ascertain whether the differences are significant, we would also use statistical tests such as ANOVA. Clearer insights would also be obtained by using boxplots or bar charts to visualize the sentiment distribution. We might look at categorization using product keywords or other available metadata if the category column is still missing.

We may use the cleaned price column that is already present in our dataset for price analysis. We may examine how attitudes differ among these groups by classifying products into price ranges (e.g., low, mid, and high price). It would be possible to ascertain whether pricing has a substantial impact on sentiment by doing statistical tests like ANOVA or t-tests. Furthermore, by managing possible outliers or log-transforming pricing values for improved distribution, we could improve our strategy. Our model, which combines textual and numerical characteristics, has already made significant strides in sentiment classification, with XGBoost attaining 95.8% accuracy. To obtain useful insights, the next steps entail adding missing category data to our dataset and doing a more thorough statistical analysis.

```
SVM Model Accuracy: 0.5527049637479086
              precision    recall  f1-score   support

           0       0.57      0.37      0.45      1761
           1       0.55      0.73      0.62      1825

    accuracy                           0.55      3586
   macro avg       0.56      0.55      0.54      3586
weighted avg       0.56      0.55      0.54      3586

/usr/local/lib/python3.11/dist-packages/xgboost/core.py:158: UserWarning: [18:46:36] WARNING: /workspace/src/learner.cc:74
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
XGBoost Model Accuracy: 0.958728388176241
              precision    recall  f1-score   support

           0       0.96      0.95      0.96      1761
           1       0.95      0.97      0.96      1825

    accuracy                           0.96      3586
   macro avg       0.96      0.96      0.96      3586
weighted avg       0.96      0.96      0.96      3586

Mean Absolute Error: 0.4434979732996923
Mean Squared Error: 0.24782552409223932
R-Squared Score: -1.089162596468185e-05
```

Analysis on results:

The findings indicate that the Support Vector Machine (SVM) and XGBoost models perform significantly differently when it comes to sentiment classification. With comparatively poor precision and recall scores, especially for negative sentiment (class 0), the SVM model's accuracy was 55.27%. The algorithm has trouble accurately identifying negative evaluations, as evidenced by the recall for negative sentiment of only 37%. Given the intricacy of textual data and the requirement for more sophisticated feature representation, this implies that the SVM model might not be a good fit for this dataset.

The XGBoost model, on the other hand, showed a significantly higher accuracy of 95.87% along with strong F1-scores, precision, and recall for both sentiment classes. This implies that XGBoost performs noticeably better at identifying patterns in the data, most likely as a result of its robust management of imbalanced data following the use of SMOTE and its capacity to manage non-linear relationships. With an R-squared score near zero (-0.00001), the linear regression model used to predict sentiment scores performed poorly despite this strong classification performance, suggesting that numerical features like price, rating, and review length are ineffective at predicting sentiment scores. This implies that literary characteristics, not numerical ones, are what largely influence sentiment. More significant numerical features could be added in the future, feature engineering could be improved.

Additional Insights from Research

The professor asked us about text processing methods we attempted beyond TF-IDF during the presentation's conclusion. Our subsequent research led us to study BERT (Bidirectional Encoder Representations from Transformers). Deep learning techniques empower BERT to comprehend contextual meanings through the analysis of word relationships in textual context, but TF-IDF evaluates text based on how often words appear. BERT helps achieve deeper understanding when analyzing product descriptions and customer queries together. The addition of BERT to future models will likely improve classification outcomes thanks to its advanced feature extraction capabilities that deliver superior text representation.

Distribution of Work

We're a highly collaborative group and try to distribute work evenly according to the members' strengths. For this project, we split the work as Junhui to analyze Question 1, Ant to analyze Question 2, and Aditya to analyze Question 3. Even though we pulled all the code together, each of us were responsible for one model: Junhui employed Random Forest, Ant used Logistic Regression, and Aditya worked with the SVM and XGBoost models. We will make necessary updates and changes to our models as we advance in the project.

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